

Oscillatory Credit Assignment:

When Phase-Relative Neural Updates Approximate Gradient Learning

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Abstract

Neural phase can organize communication, encode learning variables, and influence plasticity. None of those observations by itself solves credit assignment. This paper reconstructs a September 2024 Self Aware Networks (SAN) conjecture that treated a phase-wave differential (PWD) as a receiver-relative physical variation capable of participating in derivative-like learning. It then supplies the mathematical distinctions that the original conjecture left open. A phase departure d is separated from an objective-bearing state derivative q , a synapse-specific eligibility J , and a parameter gradient $g = J^\top q$. A calibrated phase observation can serve as a teaching estimate $\hat{q} = C_r(d)$, but the resulting update $\hat{g} = \hat{J}^\top \hat{q}$ is gradient-aligned only under explicit calibration, eligibility, timing, and error conditions.

The paper derives a two-source gradient-error bound, a relative-error alignment condition, a smooth-loss descent interval, and two impossibility results: identical phase observations can require different gradients, and a shared receiver discrepancy cannot assign different credits to different synapses without eligibility. Lean 4.32.0 checked seven scalar algebraic and logical leaves with exit code 0 and no admitted proofs. A frozen deterministic phase-event reference application produced 1,068 evidence rows, compared 12 update rules across 24 training seeds, generated six result figures and a system-context diagram, passed 18/18 validators, and reproduced all reported artifacts byte for byte.

The findings sharply bound the claim. Calibrated phase with exact eligibility was numerically equivalent to the exact gradient. Local OCA was numerically equivalent to a matched learning-signal times eligibility rule, with mean held-out accuracy 0.99992 and mean update-gradient cosine 0.80539. Phase-only learning remained at chance, phase or eligibility shuffling destroyed alignment, and a sign reversal was anti-aligned. A raw wrapped difference also solved the task, so this benchmark does not establish unique superiority of the circular calibration. Receiver-relative phase can therefore instantiate a teaching factor under a declared objective and calibration; it does not create synaptic credit by itself. The formal and application results validate the bounded model, not a cortical implementation.

Keywords: oscillatory credit assignment; phase-wave differential; gradient learning; eligibility traces; three-factor learning; biological credit assignment; formal verification; event networks

1 Introduction

Backpropagation solves a computational problem before it creates a biological one. In an artificial network, a global objective can be differentiated through a known sequence of operations, assigning each parameter a direction and magnitude of change. Biological synapses do not obviously receive an exact reverse pass,

transport symmetric weights, or retain every intervening state. Yet animals learn representations and actions across deep, recurrent, temporally extended circuits. The resulting question is not simply whether the brain uses backpropagation. It is which locally available physical variables could approximate the two mathematical ingredients that backpropagation combines: an objective-relevant teaching direction and a parameter-specific sensitivity.

Oscillatory activity is an attractive candidate substrate because phase relations can gate effective communication, carry information beyond rate, and alter plasticity. But “phase matters” is much weaker than “phase assigns credit.” A phase shift can be informative without identifying a desired target. It can alter excitability without specifying which upstream synapse should change. It can be rare under a tonic distribution without revealing the sign of a loss derivative. It can correlate with a prediction error while being epiphenomenal to the plasticity that changes behavior.

The 2024 SAN conjecture examined here was stronger than the claim that synchrony alone performs cognition and more specific than the claim that oscillations are useful clocks. It proposed a receiver-relative phase-wave differential: a bounded excitation-, inhibition-, timing-, quantity-, burst-, or omission-related departure from a tonic relation at a named receiver. That departure could be transmitted, absorbed, or used to change later Neural Tuning and functional routing [1,2]. The source then described the physical difference as representing part of a derivative-like equation in an analog biological computer. What it did not provide was a declared objective, the derivative’s units and sign, a synapse-specific credit path, an approximation theorem, or a working application.

This paper supplies that missing bridge while preserving the conjecture’s strongest form. The bridge is conditional:

receiver-relative departure

- > explicit calibration into an objective-relevant teaching estimate
- > synapse-specific eligibility
- > approximate parameter gradient
- > bounded alignment and descent conditions
- > measured plastic consequence

The negative statement is equally central. Phase departure alone has no general credit-assignment guarantee. The same observation can occur under different targets, and different synapses can require different updates under the same receiver discrepancy. The paper therefore asks not whether phase “is” a gradient, but when a phase-relative physical variable can occupy one factor in a gradient-like local update and how that claim can fail.

The contribution has four parts. First, it provides a source-faithful genealogy and refuses to treat generated dialogue as scientific validation. Second, it maps the proposal against temporal-contrast learning, phase-dependent plasticity, three-factor learning, dendritic and burst credit, and exact event gradients. Third, it derives and machine-checks bounded mathematical leaves. Fourth, it implements a controlled event-network application with matched positive, equivalence, ablation, and negative controls. The adverse result - that a raw wrapped difference performed as well as the calibrated phase derivative in the sampled task regime - is retained as a limit on the paper’s claim.

2 Source genealogy and claim ownership

2.1 September 2024 source

The earliest immutable joined source used here is `02san.md`, fixed in Git on September 6, 2024 and revised later that day [2]. A transcript of the associated video and a public September 7 article provide additional first-party context [1]. The source’s PWD is receiver-relative. It is not merely a global decrease in synchrony. A departure can be an unexpected spike, burst, inhibitory event, missing event, phase displacement, interval, or quantity change relative to a tonic relation. Its proposed significance depends on what a receiver is prepared to recognize and on what later consequence follows.

The source connects this physical difference to learning in two steps. First, the difference can alter the signals that a receiver accepts and therefore the functional route through the network. Second, I clarify that the “cost” comparison is not intended as loose metaphor. The physical variation is proposed to represent part of a derivative equation implemented by an analog biological system. This is the strongest source-faithful interpretation. It does not imply that every deviation is useful, that the network must resynchronize, or that the PWD already contains a complete gradient.

The same source prevents an overconfident reading. In that exchange, I explicitly question whether an assistant’s paraphrase has captured the conjecture and ask for atomization. Later, I request scientific papers that could validate it. A generated response then cites two oscillator studies and calls the analogy validated. This paper rejects that generated verdict. The cited studies can support component dynamics, but the source itself treats validation as unfinished.

2.2 Later SAN boundaries

Two later SAN papers own adjacent but nonidentical operations. *Inference as Training* defines a residue-and-reset criterion for distinguishing transient inference, update, persistence, and retained state change. It explicitly warns that biological feedback-driven plasticity is not ordinary backpropagation without a local rule, variables, timing, and measured residue [3]. *Phase-Wave Differential Calculus* defines the typed receiver-relative event record, conditional tonic reference, Neural Tuning update taxonomy, and adverse synthetic comparisons. It explicitly does not establish a biological gradient rule [4].

The present paper does not redefine those objects. It owns the missing credit bridge: a calibrated phase-derived teaching estimate, a synapse-specific eligibility term, an approximation bound, obstruction cases, and matched event-network tests.

2.3 Strongest-form anti-strawman reconstruction

The 2024 claim is not reduced to “desynchronization equals error.” Its strongest coherent form is:

1. an active receiver has an expected tonic relation conditioned on task and state;
2. excitation, inhibition, timing, or omission creates a bounded departure;
3. the receiver or a connected circuit can measure and transmit that difference;
4. a physical calibration maps some component of the difference into a teaching variable;
5. synapse-specific state supplies an eligibility or sensitivity term;
6. their interaction changes a route or response disposition;
7. later behavior or network state reveals whether the update was useful.

Steps 4 and 5 are necessary 2026 formalization. They are not attributed retroactively to the exact 2024 wording.

3 Prior science and the nonduplication boundary

3.1 Temporal contrast already has a gradient lineage

The idea that differences between two network states can drive useful learning is longstanding. Ackley, Hinton, and Sejnowski contrasted statistics in a Boltzmann machine [5]. O'Reilly's generalized recirculation used local activation differences to approximate error-driven learning [6]. Xie and Seung established a bounded equivalence between contrastive Hebbian learning and backpropagation in layered networks [7]. Equilibrium propagation derived an objective gradient from differences between free and weakly clamped steady states [8]. O'Reilly and colleagues later proposed a corticothalamic prediction/outcome cycle in which alpha-timed state differences drive local predictive learning [9].

In 2026, Jang and colleagues reported a particularly relevant preprint. In mouse hippocampal slices, coordinated presynaptic and postsynaptic activity increased from low to high across a 200 ms window produced long-term potentiation, high-to-low produced depression, and stable low or high conditions produced no net change [10]. The work directly tests a temporal-derivative plasticity prediction. It postdates the 2024 SAN source and strengthens the scientific basis for a physical temporal-difference term. It does not test a PWD measure, conditional tonic estimator, network objective, eligibility factor, or task-gradient alignment.

Any new paper that says only “prediction minus outcome approximates backpropagation” would duplicate this lineage.

3.2 Phase can code learning variables and control plasticity

Masquelier and colleagues showed in a model that phase-of-firing coding combined with STDP can learn repeated patterns [11]. Zanos and colleagues applied phase-locked stimulation during cortical beta oscillations in awake monkeys and observed bidirectional changes in connectivity as a function of stimulation phase [12]. Voloh and colleagues found that beta phase enhanced the coding of outcomes, reward prediction errors, and outcome history beyond phase-blind rate information for a subset of fronto-striatal spike-field pairs [13]. Madadi Asl and Ramezani Akbarabadi modeled interactions among delays, STDP, synchrony, and coupling asymmetry [14]. Borisjuk and colleagues showed that input duration can control phase response and traveling-wave behavior in an oscillator chain [15]. Lu, Guo, and Xu proposed phase-dependent plasticity for unsupervised deep networks [16].

Together these studies establish that phase can carry information, modulate effective interaction, and influence plasticity. They do not show that an arbitrary phase departure has the sign, target conditioning, or synapse specificity of a task gradient.

3.3 Three-factor and circuit credit are established mechanisms

A broad family of plasticity rules separates local pre/post eligibility from a third factor such as reward, error, novelty, or neuromodulation [17]. E-prop gives this separation a rigorous recurrent-network form: an approximate gradient is factorized into a learning signal and a synapse-local eligibility trace [18]. Feedback

alignment shows that exact backward weight symmetry is not always required for useful learning [19]. Predictive-coding networks can approach backpropagation under specified dynamics and parameters [20]. Dendritic microcircuits use compartmental error-like signals [21]. Burst-dependent plasticity and BurstCCN use bursts, feedback, short-term plasticity, and dendritic structure to coordinate hierarchical learning [22,23].

These mechanisms block any novelty claim based only on multiplying a scalar learning signal by eligibility. The residual question is whether a SAN-style conditional phase observation can be calibrated into the teaching-signal factor, what errors that introduces, and how to distinguish that role from phase as a clock, gate, or correlated readout.

3.4 Exact event gradients set a reference ceiling

EventProp derives exact gradients for a declared continuous-time leaky integrate-and-fire model [24]. Holberg and Salvi give exact pathwise-gradient results for a stochastic spiking formulation under their assumptions [25]. Such methods provide computational references. They are not automatically biologically local. In this paper, “exact gradient” always means exact for the declared reference application, not exact for cortex.

3.5 Shannon surprisal is not synaptic credit

SAN’s tonic/PWD language also invites an information-theoretic distinction. For an event d under context c , conditional self-information is

$$I(d \mid c) = -\log p(d \mid c). \quad (1)$$

A rare departure can carry more surprisal than a common tonic event [26]. But surprisal is nonnegative and does not identify whether a parameter should increase or decrease. The same surprising event can be good under one target and bad under another. Shannon information can help define unexpectedness; it cannot replace the objective derivative or eligibility factor.

Table 1 fixes the paper’s core distinctions.

Table 1: Core distinctions among phase observation, information, teaching, eligibility, and gradient.

Object	What it measures	What it does not supply by itself
tonic/PWD observation d	receiver-relative departure	desired sign, target, or synapse credit
surprisal $I(d \mid c)$	rarity under a conditional distribution	direction of loss reduction
teaching derivative q	how loss changes with receiver state	how a particular parameter changes that state
eligibility J	receiver-state sensitivity to parameters	which state direction is desired
gradient $g = J^\top q$	parameter-specific objective derivative	biological availability of its factors

4 Formal model

4.1 Objective, teaching factor, and eligibility

For bounded event or trial k , let parameters be $\theta \in \mathbb{R}^p$, receiver state be $s_k(\theta) \in \mathbb{R}^m$, and the declared loss be

$$L_k(\theta) = \ell_k(s_k(\theta)). \quad (2)$$

Define

$$q_k = \nabla_s \ell_k(s_k(\theta)), \quad J_k = \frac{\partial s_k}{\partial \theta}, \quad g_k = \nabla_\theta L_k = J_k^\top q_k. \quad (3)$$

The state derivative q_k describes the objective-relevant direction at the receiver. The Jacobian J_k assigns different sensitivities to different parameters. This is the chain-rule structure that a state difference alone lacks.

4.2 Conditional receiver-relative event

Let $\phi_{r,k}$ be a measured phase coordinate at receiver r , and let $\mu_{r,k}(c_k)$ be a conditional reference estimated from permitted pre-outcome context c_k . A circular departure can be written

$$d_{r,k} = \text{wrap}(\phi_{r,k} - \mu_{r,k}(c_k)). \quad (4)$$

The complete PWD record can also include amplitude, firing quantity, burst, inhibition, omission, interval, and uncertainty fields inherited from Paper 67 [4]. The present theorem needs only a declared observation d_k . It does not assume that d_k has the same units as q_k .

4.3 Calibration and local update

A receiver-, preparation-, objective-, and timescale-specific map calibrates the observation:

$$\hat{q}_k = C_{r,k}(d_k). \quad (5)$$

Let $\hat{J}_k$ be an eligibility estimate available to the update mechanism. The proposed gradient estimate and step are

$$\hat{g}_k = \hat{J}_k^\top \hat{q}_k, \quad \theta^+ = \theta - \eta \hat{g}_k. \quad (6)$$

Equation (6) is a three-factor structure, not a novel replacement for it. The paper's phase-specific question is whether $C_{r,k}(d_k)$ can be measured and validated as an adequate teaching estimate while other teaching routes and eligibility variables are controlled.

4.4 Biological locality is an empirical predicate

Calling a variable local requires a named implementation. A synapse might access presynaptic history, postsynaptic voltage, dendritic calcium, receptor state, neuromodulatory input, or a feedback compartment. It does not generally access an experimenter's global loss or the full Jacobian. The application therefore distinguishes exact eligibility from a fixed-feedback local proxy. Biological experiments must identify the molecular, cellular, or circuit carrier rather than infer locality from an equation.

5 Theorem ladder

5.1 Two-source approximation error

Suppressing event indices, let $g = J^\top q$ and $\hat{g} = \hat{J}^\top \hat{q}$. Add and subtract $\hat{J}^\top q$:

$$\hat{g} - g = (\hat{J} - J)^\top q + \hat{J}^\top (\hat{q} - q). \quad (7)$$

For compatible operator and vector norms,

$$\|\hat{g} - g\| \leq \|\hat{J} - J\| \|q\| + \|\hat{J}\| \|\hat{q} - q\|. \quad (8)$$

Equation (8) separates two experimental failure modes. Even perfect phase calibration cannot repair a useless eligibility path. Even exact eligibility cannot repair a teaching signal with the wrong target, sign, or timing.

5.2 Relative-error alignment

If $g \neq 0$ and

$$\|\hat{g} - g\| \leq \rho \|g\|, \quad 0 \leq \rho < 1, \quad (9)$$

then Cauchy-Schwarz gives

$$\langle g, \hat{g} \rangle = \|g\|^2 + \langle g, \hat{g} - g \rangle \geq (1 - \rho) \|g\|^2 > 0, \quad (10)$$

and $\|\hat{g}\| \leq (1 + \rho) \|g\|$. Thus $-\hat{g}$ is a descent direction before step size is chosen.

5.3 Smooth-loss descent interval

If L has beta-Lipschitz gradient, then

$$L(\theta - \eta \hat{g}) \leq L(\theta) - \eta \langle g, \hat{g} \rangle + \frac{\beta \eta^2}{2} \|\hat{g}\|^2. \quad (11)$$

Combining Equations (9)-(11) gives strict descent for nonzero (g) whenever

$$0 < \eta < \frac{2(1 - \rho)}{\beta(1 + \rho)^2}. \quad (12)$$

This is sufficient, not necessary. Useful updates can fall outside the relative-error ball, and a step can fail if its size violates the local smoothness regime.

5.4 Phase-only nonidentifiability

Suppose two target contexts have the same local observation d but require distinct gradients $g_1 \neq g_2$. Any deterministic phase-only rule $H(d)$ returns the same value in both contexts and therefore cannot equal both gradients. Target- or context-sensitive information must enter through the reference, calibration, another teaching route, or the eligibility state.

5.5 Synapse-credit obstruction

For the two-weight chain $s = w_2 w_1 x$, a shared receiver derivative q yields

$$\frac{\partial L}{\partial w_1} = q w_2 x, \quad \frac{\partial L}{\partial w_2} = q w_1 x. \quad (13)$$

The same phase-derived q cannot supply both parameter gradients without the different local sensitivities $w_2 x$ and $w_1 x$. Phase broadcast can be a learning signal; it is not the entire credit assignment.

6 Bounded formal verification

Lean 4.32.0 with Mathlib v4.32.0 checked seven leaves: a scalar factorization identity, scalar absolute-error bound, alignment from a bounded adverse error term, descent algebra from supplied hypotheses, phase-only nonidentifiability, a two-synapse obstruction, and a wrong-sign quadratic loss-increase witness. The governed check exited 0. The target source hash was 43EAFE6A62001A6A45099975BD43F2393F6BCA43BF6C5DC49ED5CE2E4DBBBB4A, and a whole-word scan found no sorry, admit, or axiom keyword.

The formal layer is intentionally bounded. It does not formalize cortical phase, a neuron, a synapse, or the empirical truth of any premise. The vector/operator-norm derivation and full smooth-loss statement remain manuscript mathematics; the checked leaves certify a scalar algebraic kernel and finite impossibility logic.

7 Reference application

7.1 System context

The application is a deterministic phase-reduced event network. Each modeled unit emits one event per oscillatory cycle. It is an executable proof object for the equations, not a conductance-based neuron model.

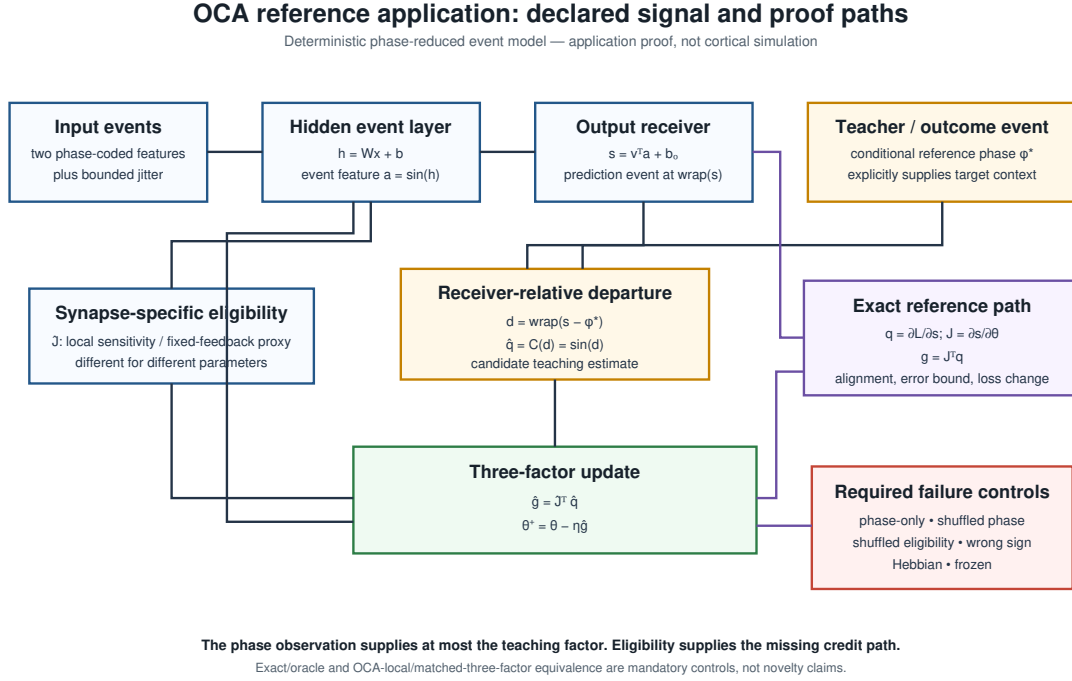


Figure 1: System context and signal paths.

Two binary variables are encoded as noisy input event phases near 0 or π . With feature vector (x_n), eight hidden event units have

$$h_n = Wx_n + b_h, \quad a_n = \sin(h_n). \quad (14)$$

Each hidden unit emits at wrapped phase $\text{wrap}(h_{nj})$; the sine value a_{nj} is the downstream readout coordinate, not an additional event count. The output receiver state is

$$s_n = v^T a_n + b_o, \quad \phi_n = \text{wrap}(s_n). \quad (15)$$

The target/outcome phase is 0 for equal input bits and π for unequal bits. The circular loss and receiver derivative are

$$L_n = 1 - \cos(s_n - \phi_n^*), \quad q_n = \sin(s_n - \phi_n^*). \quad (16)$$

The PWD observation is $d_n = \text{wrap}(\phi_n - \phi_n^*)$. Because sine is wrap-invariant, the calibrated oracle $\hat{q}_n = \sin(d_n)$ equals q_n to numerical precision. This is an explicit positive control: the teacher supplies the outcome reference. It is not evidence that unconditioned phase contains a target.

The exact event-state eligibilities are

$$\frac{\partial s_n}{\partial v_j} = \sin(h_{nj}), \quad \frac{\partial s_n}{\partial W_{ji}} = v_j \cos(h_{nj}) x_{ni}. \quad (17)$$

The local approximation replaces the hidden readout factor (v_j) with a fixed feedback proxy. This creates a controlled eligibility error while leaving the teaching estimate unchanged.

7.2 Frozen tasks

The pilot used four seeds and was excluded from reported results. It selected an eight-unit width, a 400-step training interval, learning rate 0.04, and gradient-norm cap 5.0 so that the exact reference could solve the task without starting saturated. The frozen reported configuration used:

- 24 training seeds;
- 48 training events per XOR pattern with phase jitter SD 0.18 radians;
- 128 held-out events per pattern with separately seeded jitter SD 0.24 radians;
- 40 seeds across 11 teaching/eligibility error pairs for the bound sweep;
- 100 same-observation/different-gradient context pairs;
- 30 robustness seeds across seven phase-noise levels plus global rotation;
- one code and configuration hash for every result row.

7.3 Update rules

Table 2: Frozen update rules and their roles in the reference application.

Rule	Teaching factor	Eligibility	Role
exact	exact q	exact J	analytic reference
OCA oracle	$\sin(d)$	exact J	calibration identity
matched 3-factor	exact q	local $\hat{J}$	established structural comparator
OCA local	$\sin(d)$	local $\hat{J}$	conditional OCA candidate
raw difference local	wrapped d	local $\hat{J}$	calibration-shape control
phase only	$\sin(d)$	uniform broadcast	missing-eligibility control
eligibility only	constant one	local $\hat{J}$	missing-teaching control
phase shuffled	permuted $\sin(d)$	local $\hat{J}$	pairing control
eligibility shuffled	$\sin(d)$	permuted $\hat{J}$	routing control
sign flipped	$-\sin(d)$	local $\hat{J}$	anti-alignment control
Hebbian	target-free output activity	local activity sensitivity	objective-free local control
frozen	zero	zero	no-learning control

Exact and OCA oracle were required to be equivalent. Matched three-factor and OCA local were also required to be equivalent when calibration was exact. These are nonduplication tests, not favorable discoveries.

7.4 Outputs and validation

The application emitted four seed-level CSV tables, summary JSON, six SVG result charts, a system-context diagram, and a hash manifest. An independent validator checked configuration freeze, code and artifact hashes, row counts, numerical finiteness, algebra, error bounds, equivalence pairs, task solvability, global

rotation, nonidentifiability, wrong-sign behavior, row provenance, and deterministic replay.

8 Results

8.1 Artifact and theorem validation

The reported application contained 1,068 rows: 440 bound cases, 288 training results, 100 nonidentifiability pairs, and 240 robustness/invariance cases. All 18 validators passed. A clean second run reproduced all 11 reported-run artifacts byte for byte.

The maximum factorization identity residual was 1.79×10^{-16} . The maximum numerical violation of the two-source bound was 1.18×10^{-16} . Of 440 error-sweep cases, 382 had measured relative gradient error $\rho < 1$. All 382 had positive gradient alignment and all 382 decreased loss at the frozen step size.

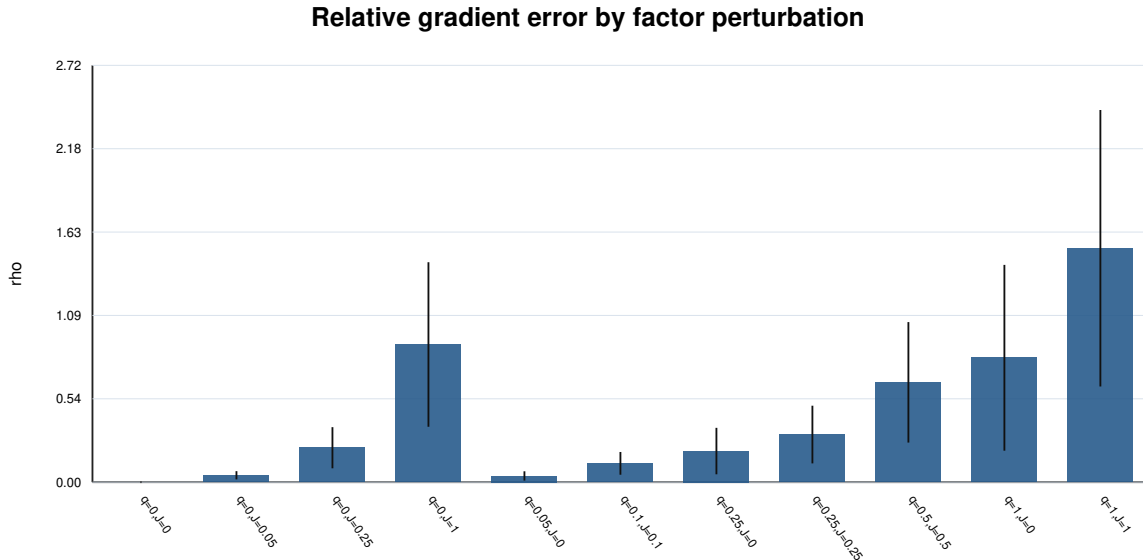


Figure 2: Relative gradient error across teaching and eligibility perturbations.

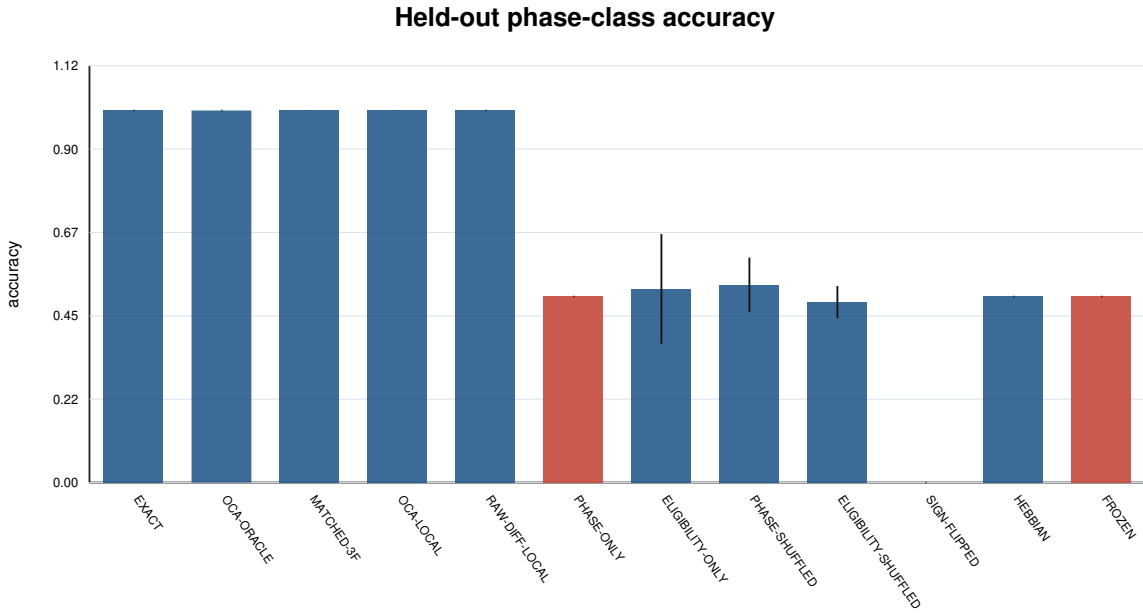
All 100 constructed context pairs had the same local phase observation and different exact gradients. A phase-only update matched both required gradients in zero cases. Common rotation of prediction and target phases changed the calibrated update by at most 3.67×10^{-16} , confirming receiver-relative rather than absolute-phase behavior in the implementation.

8.2 Training and held-out performance

Table 3 reports means and standard deviations across 24 independently initialized seeds. These are descriptive summaries of deterministic model runs, not inferential statistics about a biological population.

Table 3: Held-out performance and training-alignment summaries across 24 independently initialized seeds.

Rule	Held-out accuracy	Held-out circular loss	Mean gradient cosine	Positive-alignment fraction
exact	1.00000 +/- 0.00000	0.00608 +/- 0.00193	1.00000	1.00000
OCA oracle	1.00000 +/- 0.00000	0.00608 +/- 0.00193	1.00000	1.00000
matched 3-factor	0.99992 +/- 0.00039	0.01167 +/- 0.00858	0.80539	1.00000
OCA local	0.99992 +/- 0.00039	0.01167 +/- 0.00858	0.80539	1.00000
raw difference local	1.00000 +/- 0.00000	0.00927 +/- 0.00394	0.76490	0.99844
phase only	0.50000 +/- 0.00000	0.98815 +/- 0.02476	0.01690	0.49135
eligibility only	0.52067 +/- 0.14733	0.96469 +/- 0.21014	-0.00149	0.49479
phase shuffled	0.53133 +/- 0.07337	0.96773 +/- 0.04442	0.06763	0.58177
eligibility shuffled	0.48511 +/- 0.04320	1.02516 +/- 0.06672	-0.02234	0.39844
sign flipped	0.00000 +/- 0.00000	1.98828 +/- 0.00615	-0.83197	0.00000
Hebbian	0.50000 +/- 0.00000	1.00001 +/- 0.00009	-0.00238	0.52010
frozen	0.50000 +/- 0.00000	0.99230 +/- 0.02038	0.00000	0.00000

**Figure 3:** Held-out phase-class accuracy across update rules.

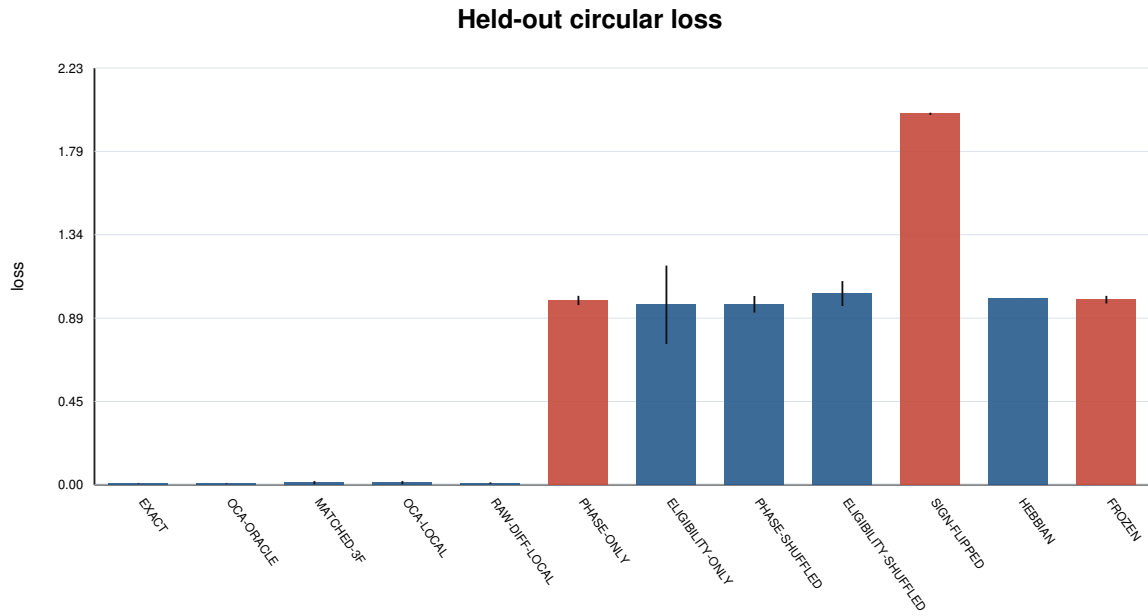


Figure 4: Held-out circular loss across update rules.

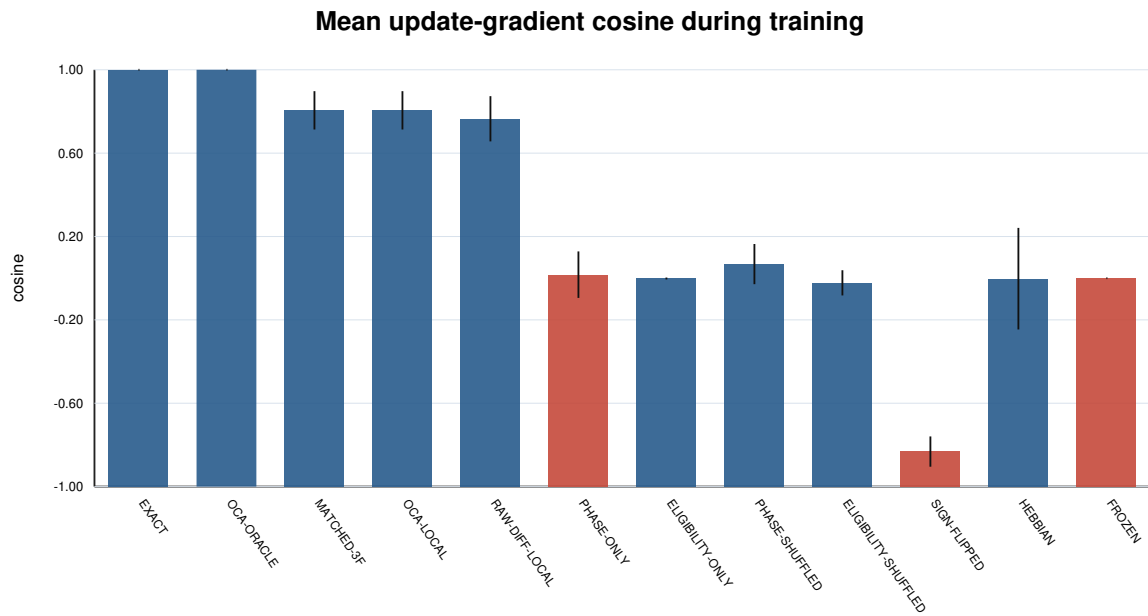


Figure 5: Mean update-gradient cosine during training.



Figure 6: Fraction of training steps that decreased loss.

The maximum paired held-out loss difference between exact and OCA oracle was 1.04×10^{-17} ; the maximum accuracy difference was zero. The maximum paired loss difference between matched three-factor and OCA local was 3.47×10^{-17} ; the maximum accuracy difference was zero. These equivalences are the expected result when the phase calibration exactly reproduces the learning signal used by the comparator.

Phase-only stayed at chance. This control used one uniform batch-broadcast update, so balanced full-batch observations can cancel its mean teaching factor; it is a specific missing-eligibility countermodel, not a proof that every phase-dependent local rule fails. Phase shuffling and eligibility shuffling also prevented reliable learning. Sign reversal produced systematic anti-alignment and zero held-out accuracy. The target-free Hebbian rule did not solve the declared XOR objective.

8.3 Adverse calibration result

The raw wrapped difference plus local eligibility reached perfect mean held-out accuracy and a lower mean loss than OCA local, although its mean gradient cosine was smaller. This is adverse to a claim that the sine calibration is uniquely necessary in the sampled regime. The raw difference and sine have the same sign and similar magnitude near zero. The learned trajectories evidently remained in a region where that monotone approximation was adequate.

The correct conclusion is therefore mathematical, not comparative: sine is the exact receiver-state derivative for the declared circular loss and is invariant to wrapping. The application does not establish that biological circuits must use sine or that this calibration beats every local monotone alternative.

8.4 Phase-noise robustness

Teaching-phase noise degraded gradient alignment continuously. Mean cosine was approximately 1.00, 0.99, 0.98, 0.89, 0.72, 0.58, and 0.49 at noise SD 0, 0.05, 0.10, 0.25, 0.50, 0.75, and 1.00 radians. Mean relative error crossed approximately one between 0.50 and 0.75 radians in this configuration.

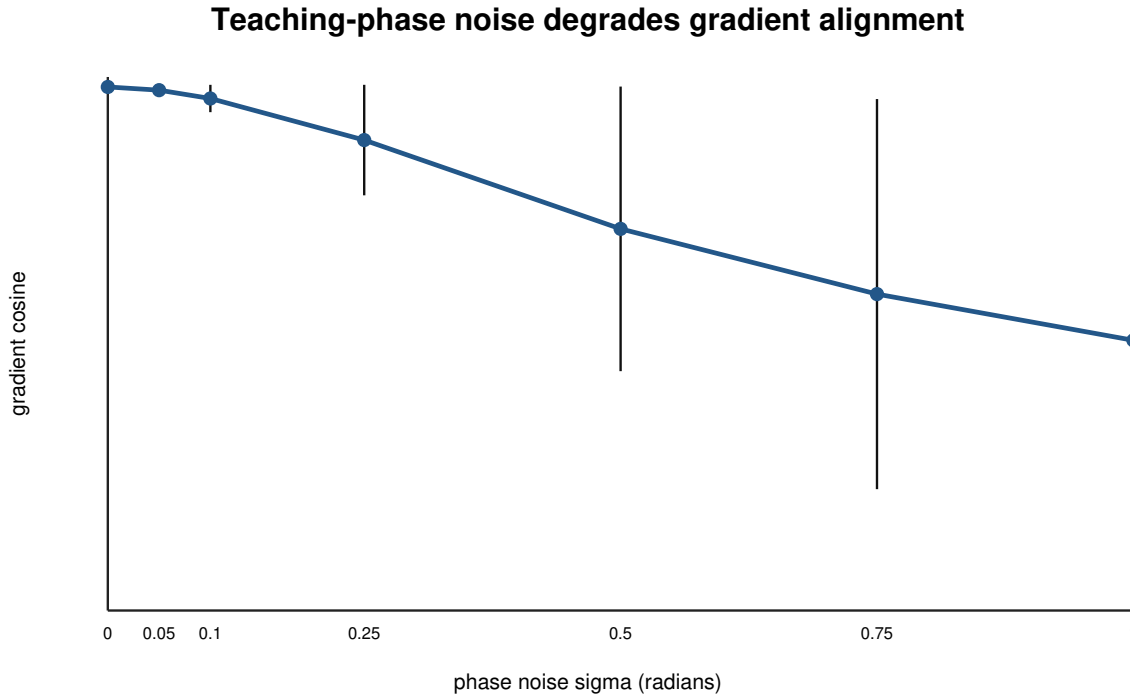


Figure 7: Teaching-phase noise and gradient alignment.

This is a model-specific calibration curve. It is not a physiological noise threshold.

9 Interpretation

9.1 What the application establishes

The application establishes five things inside its declared model.

First, a receiver-relative phase difference can exactly encode a teaching derivative when the objective and calibration make that true. Second, the phase-derived teaching factor remains incomplete without parameter-specific eligibility. Third, errors in teaching and eligibility contribute separately to gradient error, and the measured error bound held numerically. Fourth, relative gradient error below one predicted positive alignment and, under the frozen step, loss decrease in every tested case. Fifth, phase-only, shuffled-route, same-observation/different-gradient, and wrong-sign controls failed in the predicted direction.

9.2 What it does not establish

The application does not show that cortex calculates the declared circular loss. It does not show that a tonic neural state is a target phase. It does not show that a biological synapse has exact or fixed-feedback eligibility. It does not identify a molecular calibration C_r . It does not show that phase is better than voltage, rate, dendritic calcium, burst probability, or another variable after matched controls. It does not show SAN-specific algorithmic superiority.

The strongest computational conclusion is a conditional equivalence: when phase provides an adequate teaching signal and eligibility provides the credit path, OCA instantiates the established three-factor form. The strongest negative conclusion is an impossibility: without target/context conditioning or eligibility, phase alone cannot generally assign credit.

9.3 Physical representation is not semantic magic

The 2024 analog-computer intuition survives in a more precise form. A voltage, phase offset, burst probability, or concentration can physically represent a mathematical factor when the circuit's input-output relations implement a stable calibration and when downstream plasticity uses that factor. The variable does not acquire derivative semantics merely by changing. The semantics come from the declared objective relation, receiver calibration, eligibility path, and causal consequence.

9.4 Synchrony, difference, and correction

The result also clarifies the role of synchrony. A tonic or coherent relation can provide a conditional reference and a communication window. A PWD can differentiate an event relative to that reference. But the desired update is not always resynchronization. If the target calls for a stable phase lag, alternating route, or learned desynchronization, reducing the departure toward zero could be wrong. Correction means reducing a declared objective under the relevant context, not maximizing synchrony in the abstract.

10 Biological research program

The next experiments should test the factorization rather than search for a single oscillatory signature of learning.

10.1 Measurement design

A strong study would combine:

1. a named receiver population and presynaptic pathway;
2. pre-outcome estimation of a conditional tonic phase relation;
3. trial-level phase, rate, power, burst, inhibition, and timing measures;
4. a declared objective-relevant variable such as prediction error or behavioral correction;
5. a candidate local eligibility measure, such as pre/post timing, dendritic voltage, calcium, receptor state, or synaptic tag;
6. later synaptic efficacy or route-change measurement;

7. phase-specific perturbation that matches rate, power, movement, and stimulation energy;
8. held-out calibration and preregistered negative controls.

10.2 Decisive comparisons

The central model comparison should ask whether

phase-derived teaching estimate + eligibility

predicts later plasticity better than each of:

phase alone

eligibility alone

rate/amplitude + eligibility

burst + eligibility

unconditioned phase

phase with receiver labels shuffled

phase with outcome labels shuffled

matched nonoscillatory state difference + eligibility

If phase adds no held-out value after matched state variables, the phase-specific claim narrows. If the phase term works only as a clock or gate, it should be reported as timing rather than credit.

10.3 Falsifiable predictions

Table 4: Falsifiable predictions, current supporting results, and decisive failure conditions.

Prediction	Supporting result	Falsifier
same local phase can require different updates under different targets	nonidentifiability theorem and 100 paired contexts	phase-only predicts both target-conditioned gradients without hidden target leakage
eligibility is required for synapse-specific credit	chain-rule theorem and shuffled-eligibility failure	uniform phase broadcast predicts parameter-specific updates as well as measured eligibility under matched controls
teaching and eligibility errors transfer separately	Equation (8) and 440-case sweep	measured gradient error systematically exceeds the declared bound after assumptions and numerical error are verified
wrong sign can increase loss	Lean witness and sign-flipped control	sign reversal preserves alignment and outcome under an objective for which sign is identified
relative phase should be invariant to common rotation	Equation (4) and rotation validator	common reference/prediction rotation changes the calibrated update without another absolute-phase variable

Prediction	Supporting result	Falsifier
calibration advantage depends on operating regime	raw-difference adverse result	one calibration dominates across wrap boundaries, targets, noise, and preparations under preregistered matched tests
OCA should collapse to matched three-factor learning when signals match	numerical equivalence	identical q and $\hat{J}$ produce materially different updates after implementation error is excluded

10.4 Tissue-level bridge

The Jang et al. slice result provides a candidate temporal-plasticity component [10]. Zanos et al. provide causal phase/plasticity evidence in awake cortex [12]. Voloh et al. provide phase coding of learning variables [13]. A decisive OCA experiment would join these scales without assuming their combination: measure an outcome-conditioned phase signal, a local eligibility state, and a subsequent synaptic or behavioral update in the same preparation; then perturb phase while controlling alternative variables.

11 Limitations

First, the reference application is phase-reduced and feedforward. It emits events but does not simulate ion channels, conductances, dendritic morphology, recurrent cortical dynamics, or neuromodulation. A recurrent LIF or conductance-based extension should test whether the conclusions survive spike discontinuities, delays, and state-dependent eligibility.

Second, the target/outcome phase is explicitly supplied. This makes the positive control transparent but does not solve the biological question of where target context comes from. Predictive, reinforcement, homeostatic, or action-feedback circuits would provide different teaching routes.

Third, the fixed-feedback proxy is only one eligibility approximation. Dendritic compartments, e-prop traces, burst multiplexing, and learned feedback offer stronger comparators. Future applications should implement them under matched capacity.

Fourth, the XOR task is deliberately small. It exposes algebra and negative controls but cannot establish scalability, continual learning, robustness to distribution shift, or biological efficiency.

Fifth, the theorem’s relative-error condition is sufficient and global in parameter norm. The error sweep injected controlled synthetic teaching and eligibility perturbations; it did not estimate those errors from tissue. The application measured loss decrease at the frozen step but did not estimate a beta-smoothness constant or numerically certify Equation (12)’s step-size interval. A useful biological rule may rely on average alignment, subspace alignment, stochastic descent, or task-specific geometry even when $\rho \geq 1$.

Sixth, the biological literature supports components at different species, regions, frequencies, preparations, and timescales. Their conjunction is a research program, not an already observed mechanism.

Seventh, the source genealogy establishes priority only for the exact archived SAN wording. It does not displace earlier temporal-contrast, phase-plasticity, or three-factor learning work, all of which are credited here.

12 Conclusion

A neural phase departure can be meaningful without being a gradient. It becomes part of a gradient-like learning update only when a declared circuit and objective calibrate it into an adequate teaching factor and combine it with synapse-specific eligibility. Under bounded factor errors, the resulting update has a bounded gradient error; under a relative-error condition and an appropriate step, it is a descent direction. Without target conditioning or eligibility, phase-only credit is nonidentifiable.

The formal and application results support that conditional statement. They do not support the slogan that PWD is biological backpropagation. In the reference model, calibrated phase plus exact eligibility equaled the exact gradient; calibrated phase plus local eligibility equaled a matched three-factor rule; phase alone failed; and wrong-sign or shuffled factors failed. A raw wrapped difference also worked, preventing a claim of unique calibration superiority.

The resulting SAN contribution is narrower and stronger: a receiver-relative phase-wave differential is a candidate physical teaching observation whose computational meaning must be earned by calibration, eligibility, and consequence. The experiments that matter next are those that measure all three in the same biological circuit.

Data, code, and formal materials

The companion package contains the exact source ledger, primary-source ledger, nonduplication map, Lean source and check receipt, frozen experiment specification, Python/NumPy application, configuration, 1,068-row result tables, six result figures, architecture diagram, validation report, and deterministic-run manifests. Public provider links will be added only after an exact release package is authorized and deposited.

Author contributions

Micah Blumberg conceived the SAN conjecture, directed the source reconstruction, specified the strongest-form interpretation and research goal, and is the sole author responsible for the paper's claims. AI systems assisted with source routing, literature discovery, mathematical formalization, code implementation, validation, figure production, and editorial drafting under author direction. Generated dialogue from the 2024 source is not treated as independent evidence or coauthorship.

Competing interests

The author declares no competing financial interest. Self Aware Networks and phase-wave differential are terms used in the author's research program. This preprint has not been peer reviewed.

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